Understanding a Probabilistic Approach to Solving Ordinary Differential Equations

A Detailed Exposition

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Abstract

This document provides a detailed explanation of concepts presented in a research paper on probabilistic numerical methods for solving Initial Value Problems (IVPs) of Ordinary Differential Equations (ODEs). We begin by defining ODEs and the traditional approach to solving them, then delve into the probabilistic framework where the ODE solution is treated as a quantity to be inferred, thereby allowing for principled uncertainty quantification. Key components such as the prior model (based on Stochastic Differential Equations), the filtering algorithm (akin to a Kalman filter), the role of measurement noise, and the concept of state misalignment are thoroughly discussed. Finally, we examine the local error analysis of such a filter, including the necessary assumptions and key theorems establishing convergence properties. All mathematical notation is explained in context, aiming for clarity for an undergraduate reader with a background in calculus, linear algebra, and basic probability.

1 Introduction to ODEs and the Problem Setting

The purpose of this section is to introduce the fundamental mathematical problem addressed by the paper: solving Ordinary Differential Equations.

An Ordinary Differential Equation (ODE) is a mathematical equation that describes how a quantity or a set of quantities change with respect to a single independent variable, which is typically time. An Initial Value Problem (IVP) for an ODE provides not only the equation governing the change but also the specific state of the system at an initial point in time. This initial state is necessary to single out a unique solution from the family of possible solutions to the ODE.

The paper considers an **autonomous** ODE. An autonomous system is one where the rules governing its evolution (i.e., the rate of change) depend only on its current state, not explicitly on the independent variable (time in this case). The specific form of the IVP is:

$$x^{(1)}(t) := \frac{dx}{dt}(t) = f(x(t)), \quad \forall t \in [0, T] \tag{1}$$

with an initial condition:

$$x(0) = x_0 \in \mathbb{R}^d. \tag{2}$$

Let's break down the notation:

- t: The independent variable, representing time. The solution is sought over a finite interval $[0, T]$, where $T > 0$ is the final time.
- $x(t) \in \mathbb{R}^d$: This is the **state vector** of the system at time t. It is a d-dimensional vector, meaning the system can be described by d different, possibly coupled, variables. For notational convenience, $x(t)$ is sometimes referred to as $x^{(0)}(t)$ (the 0-th derivative).

- $x^{(1)}(t)$: This denotes the first time derivative of the state vector $x(t)$, i.e., $\frac{dx}{dt}(t)$. It represents the instantaneous rate of change of the system's state.
- $f : \mathbb{R}^d \to \mathbb{R}^d$: This is a given function, often called the **vector field** or dynamics function. It maps the current state $x(t)$ to its rate of change $x^{(1)}(t)$. The function f encapsulates the underlying laws governing the system.
- $x_0 \in \mathbb{R}^d$: This is the known **initial state vector** at the starting time $t = 0$.

The primary objective of an ODE solver is to find an **approximate solution**, denoted $\hat{x}(t)$ (or often $m(t)$ in the probabilistic context of the paper), to the true, but often analytically unobtainable, solution $x(t)$ over the time interval $[0, T]$. Traditional numerical solvers (like Euler's method or Runge-Kutta methods) typically provide a sequence of point estimates for $x(t)$ at discrete time steps.

The research paper discussed here adopts a **probabilistic perspective**. Instead of just computing a single approximate solution, this approach aims to quantify the uncertainty associated with the numerical solution. The numerical solution $\hat{x}$ (referred to as the *estimator*) of the exact, unknown solution x (the *estimand*) is treated as a random variable or, more accurately, a random process. The ODE solver itself is viewed as an inference algorithm that produces a probability distribution over possible solutions, reflecting the accumulated uncertainty from discretization and other potential error sources. This is a departure from classical methods, offering a richer output that includes error estimates as a natural byproduct.

2 Probabilistic Modeling of ODE Solutions

Traditional numerical solvers for ordinary differential equations (ODEs) provide deterministic point estimates for solutions, without accounting for uncertainty. In contrast, **probabilistic numerics** treats the solution of an ODE as a *random function* and solves it using tools from stochastic processes. In this context, the solution trajectory is modeled using Gaussian processes with specific structural properties. This section explains three such key processes: **Integrated Brownian Motion (IBM)**, **Integrated Ornstein-Uhlenbeck Process (IOUP)**, and the **Gauss-Markov Process**, which serve as priors in probabilistic ODE solvers.

2.1 Gauss-Markov Process

Definition

A **Gauss-Markov process** is a stochastic process that satisfies:

- It is a **Gaussian process** (i.e., any finite collection of random variables has a multivariate normal distribution),
- It has the **Markov property**, i.e., the future is conditionally independent of the past given the present.

Mathematical Formulation

Let $\{X(t)\}_{t\in\mathbb{R}}$ be a real-valued stochastic process. Then $X(t)$ is a Gauss-Markov process if:

$$\mathbb{P}(X(t_{n+1}) \mid X(t_n), X(t_{n-1}), \ldots, X(t_0)) = \mathbb{P}(X(t_{n+1}) \mid X(t_n))$$

and all finite-dimensional distributions are Gaussian.

Use in ODE Solvers

In probabilistic ODE solvers, Gauss-Markov processes allow efficient Bayesian filtering or smoothing methods (such as Kalman filters) to propagate distributions over the solution of an ODE.

2.2 Integrated Brownian Motion (IBM)

Definition

Let $B(t)$ be standard Brownian motion with:

$$\mathbb{E}[B(t)] = 0, \quad \mathrm{Cov}(B(s), B(t)) = \min(s, t)$$

The n-times integrated Brownian motion (IBM) is defined as:

$$X_n(t) = \int_0^t \int_0^{t_n} \cdots \int_0^{t_2} B(t_1)\, dt_1\, dt_2 \cdots dt_n$$

For example, the first-order IBM is:

$$X_1(t) = \int_0^t B(s) ds$$

Properties

- IBM is a smooth Gaussian process.
- It has zero mean and a well-defined covariance structure.
- The n-th derivative of an n-times integrated Brownian motion is Brownian motion itself.

Covariance Function

For first-order IBM:

$$\mathrm{Cov}(X_1(s), X_1(t)) = \int_0^s \int_0^t \min(u, v)\, du\, dv$$

Application in Probabilistic ODEs

When solving ODEs of the form:

$$\frac{dy}{dt} = f(t, y(t)), \quad y(t_0) = y_0$$

we model $y(t)$ as an IBM:

$$y(t) \sim \mathcal{GP}(m(t), k(t, t'))$$

and condition this prior on evaluations of f to get a posterior distribution over the solution.

2.3 Integrated Ornstein-Uhlenbeck Process (IOUP)

Ornstein-Uhlenbeck (OU) Process

The OU process is a mean-reverting stochastic process defined by the SDE:

$$dX_t = -\theta X_t\, dt + \sigma\, dW_t$$

where:

- $\theta > 0$ controls the rate of mean reversion,
- σ is the diffusion coefficient,
- W_t is standard Brownian motion.

IOUP Definition

The **Integrated Ornstein-Uhlenbeck Process (IOUP)** is defined as:

$$Y(t) = \int_0^t X(s)\, ds$$

where $X(s)$ is an OU process.

Properties

- IOUP models functions with bounded growth and damping.
- It has a smooth sample path (more realistic than IBM for some physical systems).
- Suitable for modeling decaying or stabilizing dynamics.

Covariance Structure

The OU process has stationary covariance:

$$\mathrm{Cov}(X(s), X(t)) = \frac{\sigma^2}{2\theta} e^{-\theta|t-s|}$$

From this, the covariance of IOUP can be derived via double integration.

Application in Probabilistic ODEs

Like IBM, IOUP can be used as a prior over the solution $y(t)$:

$$y(t) \sim \mathcal{GP}(m(t), k_{\mathrm{IOUP}}(t, t'))$$

This is useful for problems where the true solution is known to decay or be bounded.

2.4 Comparison of IBM and IOUP

Property	**IBM**	**IOUP**
Smoothness	Very smooth	Smooth and decaying
Covariance growth	Unbounded (grows with t)	Bounded
Mean behavior	Constant or linear	Mean-reverting
Suitability	General ODEs	Systems with damping or decay

Table 1: Comparison between IBM and IOUP as priors in ODE solvers

2.5 Conclusion

Probabilistic solvers for ODEs offer a principled way to quantify uncertainty in the numerical solution of differential equations. Key to these solvers are Gaussian priors over the solution space, such as:

- **Gauss-Markov processes** for tractable inference,
- **Integrated Brownian Motion (IBM)** for smooth, unbounded trajectories,
- **Integrated Ornstein-Uhlenbeck Process (IOUP)** for decaying or bounded systems.

These models can be embedded in a Bayesian filtering framework (e.g., using Kalman filters) to compute posterior distributions over the ODE solution at each time step.

3 The Probabilistic ODE Solver Framework

This section introduces the core methodology of the probabilistic ODE solver, which reinterprets the task of solving an IVP as a problem of statistical inference.

The central idea is to **recast the IVP as a stochastic filtering problem**. This is a powerful paradigm borrowed from signal processing and control theory, where one aims to estimate the true underlying state of a dynamic system based on noisy or incomplete measurements. In our context, the "system" is the true solution of the ODE, and the "measurements" will be derived from the ODE function f. This approach involves two main components:

1. Defining a **prior belief** about the solution $x(t)$ before incorporating the specific ODE dynamics.

2. Using an **iterative filtering algorithm** (similar to a Kalman filter) to update this prior belief as information from the ODE f becomes available at successive time points.

3.1 Prior on the Solution x

This subsection explains how an initial model of belief about the ODE solution is constructed, even before using the ODE function f for updates.

Instead of assuming nothing about the solution $x(t)$ beyond the ODE itself, the probabilistic approach begins by placing a **prior probability distribution** on it. This prior represents initial assumptions about the general behavior of the solution, such as its smoothness. For reasons of mathematical tractability and to enable efficient computation, this prior is typically chosen to be a **Gauss-Markov process**.

- A **Gaussian process** (GP) is a collection of random variables, any finite number of which have a joint Gaussian (normal) distribution. Using a GP prior is convenient because if the "measurements" are also assumed to be Gaussian (or linearly related to Gaussian states with Gaussian noise), then the updated belief (posterior) remains Gaussian.

- A **Markov process** is one where the future state depends only on the current state, not on the entire past history that led to the current state (given the current state). This "memoryless" property significantly simplifies the update calculations, allowing for recursive estimation.

A common way to define such a Gauss-Markov process $X(t)$ is as the solution to a linear Stochastic Differential Equation (SDE).

The state vector $X(t)$ in this probabilistic model is augmented to include not just the solution $x(t)$ itself (the 0-th derivative) but also its first q time derivatives, where $q \in \mathbb{N}$ is a chosen integer (e.g., $q = 1, 2, 3, \dots$). Modeling higher derivatives in the prior helps to enforce smoothness assumptions on the solution. For each dimension $j \in \{1, \dots, d\}$ of the original d-dimensional ODE, an augmented state vector $X_j(t)$ is defined as:

$$X_j(t) = \begin{pmatrix} X_j^{(0)}(t) \\ X_j^{(1)}(t) \\ \vdots \\ X_j^{(q)}(t) \end{pmatrix} \in \mathbb{R}^{q+1}, \tag{3}$$

where $X_j^{(i)}(t)$ represents the random process modeling the i-th time derivative of the j-th component of the true solution $x(t)$. Thus, $X_j^{(0)}(t)$ models the j-th component of $x(t)$.

This $(q+1)$-dimensional stochastic process $X_j(t)$ is assumed to evolve according to the following linear SDE (Equation (2) in the paper):

$$dX_j(t) = \underbrace{\begin{pmatrix} 0 & 1 & 0 & \dots & 0 \\ \vdots & \ddots & \ddots & \ddots & \vdots \\ 0 & \dots & 0 & 1 & 0 \\ 0 & \dots & \dots & 0 & 1 \\ c_0 & c_1 & \dots & c_{q-1} & c_q \end{pmatrix}}_{F_j \text{ (Drift Matrix)}} X_j(t)dt + \underbrace{\begin{pmatrix} 0 \\ \vdots \\ 0 \\ \sigma_j \end{pmatrix}}_{L_j \text{ (Diffusion Vector)}} dB_j(t). \tag{4}$$

Let's dissect this SDE:

- $dX_j(t)$: Represents the infinitesimal change in the state vector $X_j(t)$ over an infinitesimal time interval dt.
- F_j: This is the **drift matrix**. It governs the deterministic part of the evolution of $X_j(t)$.
  - The first q rows of F_j (with 1s on the super-diagonal) enforce the kinematic relationships: $dX_j^{(0)}/dt = X_j^{(1)}(t)$, $dX_j^{(1)}/dt = X_j^{(2)}(t)$, ..., $dX_j^{(q-1)}/dt = X_j^{(q)}(t)$. These simply state that the derivative of the $(i-1)$-th derivative is the i-th derivative.
  - The last row, with coefficients $(c_0, \dots, c_q)$, defines the specific dynamics assumed for the highest modeled derivative, $X_j^{(q)}(t)$: $dX_j^{(q)}/dt = c_0 X_j^{(0)}(t) + \dots + c_q X_j^{(q)}(t) + \text{noise}$. The choice of these c_k coefficients determines the type of prior (e.g., IBM or IOUP, discussed below).
- L_j: This is the **diffusion vector**. It specifies how the random noise component affects the state. In this model, the noise (scaled by σ_j) directly influences only the highest derivative $X_j^{(q)}(t)$. The idea is that randomness enters at the highest level of dynamics, and its effect propagates to lower derivatives through integration.
- $dB_j(t)$: This is an infinitesimal increment of a one-dimensional standard **Brownian motion** (also known as a Wiener process) for the j-th component. Brownian motion is a continuous-time stochastic process often used to model random fluctuations. $\sigma_j > 0$ is a scaling factor for the noise intensity. The Brownian motions $B_j(t)$ for different components j (i.e., for different dimensions of the original ODE) are assumed to be mutually independent.
- The initial condition for this SDE is also probabilistic: $X_j(0) \sim \mathcal{N}(m_j(0), P_j(0))$. This means $X_j(0)$ is drawn from a Gaussian (Normal) distribution with a specified mean vector $m_j(0) \in \mathbb{R}^{q+1}$ and covariance matrix $P_j(0) \in \mathbb{R}^{(q+1)\times(q+1)}$. The initial states $X_j(0)$ for different j are also assumed to be independent.

The solution to a linear SDE like Eq. (4) with a Gaussian initial condition is indeed a Gauss-Markov process. The mean $m_j(t)$ and covariance $P_j(t)$ of this process $X_j(t)$ at any time $t \geq 0$ (given $m_j(0)$ and $P_j(0)$) propagate according to:

$$m_j(t) = A_j(t) m_j(0) \tag{5}$$

$$P_j(t) = A_j(t) P_j(0) A_j(t)^T + Q_j(t). \tag{6}$$

Here:

- $A_j(t)$ is the **state transition matrix** for the interval $[0, t]$. It describes how the mean of the state evolves deterministically. For an autonomous (time-invariant) drift matrix F_j, $A_j(t) = \exp(F_j t)$, where $\exp(\cdot)$ denotes the matrix exponential.

- $Q_j(t)$ is the **process noise covariance matrix** accumulated over the interval $[0, t]$. It quantifies the uncertainty introduced by the Wiener process term $L_j dB_j(t)$ during this interval. It is calculated as $Q_j(t) = \int_0^t A_j(t-s) L_j L_j^T A_j(t-s)^T ds$.

The paper under discussion focuses on a specific choice for the coefficients in the last row of F_j: $(c_0, \ldots, c_q) := (0, \ldots, 0, -\theta)$ for some parameter $\theta \geq 0$ (Equation (5) in the original paper). This choice leads to two common types of priors:

- If $\theta = 0$: All c_k are zero. This corresponds to the q-times **Integrated Brownian Motion (IBM)** prior. In this model, the q-th derivative $X_j^{(q)}(t)$ behaves like a Wiener process (its derivative is white noise). This prior is very flexible and imposes minimal assumptions beyond smoothness.

- If $\theta > 0$: $c_q = -\theta$ and other $c_k = 0$. This corresponds to the q-times **Integrated Ornstein-Uhlenbeck Process (IOUP)** prior. The q-th derivative $X_j^{(q)}(t)$ behaves like an Ornstein-Uhlenbeck process, which is characterized by its tendency to revert to a mean (in this case, zero). θ is the drift or mean-reversion parameter; a larger θ implies stronger reversion. IOUP priors often lead to smoother and more "well-behaved" sample paths than IBM priors.

For these IBM and IOUP priors, the matrices $A(t)$ and $Q(t)$ have known analytical closed-form expressions (given by Eqs. (6) and (7) in the original paper). These expressions involve t, θ, σ_j, and factorials, reflecting the integration of the underlying SDE dynamics.

For simplicity in describing the subsequent filtering algorithm, the paper often assumes $d = 1$ (a scalar ODE, meaning $x(t)$ is a single value rather than a vector). In this case, the state vector $X(t)$ (dropping the subscript j) is still $(q+1)$-dimensional, representing the scalar solution and its first q derivatives. The extension to $d > 1$ is conceptually straightforward: due to the assumed independence of the $X_j(t)$ processes in the prior, the same filtering logic can be applied independently to each of the d components.

3.2 The Filtering Algorithm (Section 2.2)

This subsection describes the iterative process by which the prior belief about the solution is updated using information from the ODE.

The ODE filter algorithm, denoted Ψ in the paper, iteratively computes the **filtering mean** $m(nh)$ and **filtering covariance** $P(nh)$ of the augmented state $X(nh)$ at discrete time steps $t_n = nh$. Here, h is the step size. This algorithm is a specific instance of a **Kalman filter**, tailored to the SDE prior (Eq. (4)) and a measurement model derived from the ODE f.

Each step of the filter, advancing the solution from time $t = nh$ to $t' = (n+1)h = t + h$, consists of two main phases: **Prediction** and **Update**.

Let $m(t)$ and $P(t)$ denote the current mean and covariance of the state $X(t)$ at time $t = nh$.

1. Prediction Step (Time Update): In this phase, the filter uses its knowledge of the system dynamics (from the prior SDE, Eq. (4)) to predict the state at the next time point $t' = t + h$, before incorporating any new "measurement" information from the ODE f. The prediction equations are derived directly from Eqs. (5) and (6) applied over a single time interval of length h:

$$m^-(t+h) := A(h)m(t) \tag{7}$$

$$P^-(t+h) := A(h)P(t)A(h)^T + Q(h) \tag{8}$$

Here:

- $m^-(t+h) \in \mathbb{R}^{q+1}$ is the **predicted mean** (also called the *a priori* mean) of the state $X(t+h)$. The superscript " denotes a quantity *before* the measurement update.

- $P^-(t+h) \in \mathbb{R}^{(q+1)\times(q+1)}$ is the **predicted covariance** (or *a priori* covariance), reflecting the uncertainty in $m^-(t+h)$.

- $A(h)$ and $Q(h)$ are the state transition matrix and process noise covariance matrix, respectively, for a single time interval of length h. These are derived from the chosen SDE prior (IBM or IOUP).

The components of the predicted mean vector $m^-(t+h)$ are $m^{-(i)}(t+h)$ for $i = 0, \ldots, q$, representing the predicted solution value and its predicted derivatives.

2. Update Step (Measurement Update): In this phase, the filter incorporates information derived from the ODE, $x^{(1)}(t) = f(x^{(0)}(t))$, to refine its prediction. This ODE relationship is treated as providing a "measurement" related to the first derivative of the solution.

- **Measurement Model**: The filter obtains a "measurement" $y(t+h)$ by evaluating the ODE function f using its current best guess for the solution value $x^{(0)}(t+h)$, which is the predicted value $m^{-(0)}(t+h)$:

$$y(t+h) := f(m^{-(0)}(t+h)). \tag{9}$$

  For a d-dimensional ODE, $y(t+h) \in \mathbb{R}^d$. If $d = 1$ (scalar ODE), then $y(t+h) \in \mathbb{R}$. This $y(t+h)$ is interpreted as an observation of the true first derivative $x^{(1)}(t+h)$, because if $m^{-(0)}(t+h)$ were equal to the true $x^{(0)}(t+h)$, then $f(m^{-(0)}(t+h))$ would be the true $x^{(1)}(t+h)$. Since $m^{-(0)}(t+h)$ is only an estimate, $y(t+h)$ is a "noisy" observation of $x^{(1)}(t+h)$.

- **Innovation (or Residual)**: The innovation represents the new information provided by the measurement. It's the difference between the actual measurement $y(t+h)$ and what the filter *predicted* that measurement would be. The filter predicted the first derivative to be $m^{-(1)}(t+h)$ (the second component of $m^-(t+h)$). So, the innovation is:

$$r(t+h) := y(t+h) - m^{-(1)}(t+h). \tag{10}$$

  For $d = 1$, $r(t+h) \in \mathbb{R}$. If $r(t+h)$ is large, it means the ODE f is suggesting a first derivative that is significantly different from what the prior SDE predicted.

- **Measurement Noise Variance** $R(t+h)$: This crucial parameter quantifies the uncertainty associated with the measurement $y(t+h)$. For $d = 1$, $R(t+h)$ is a scalar variance. It reflects how much $y(t+h) = f(m^{-(0)}(t+h))$ is expected to deviate from the true derivative $\dot{x}(t+h) = f(x^{(0)}(t+h))$ due to the error in $m^{-(0)}(t+h)$ and any other uncertainties in f. The choice of R is discussed further in Section 3.3.

- **Kalman Gain** $\beta(t+h)$: The Kalman gain determines how much the predicted state estimate $m^-(t+h)$ should be adjusted in response to the innovation $r(t+h)$. It optimally balances the confidence in the prediction (from $P^-(t+h)$) with the confidence in the measurement (related to $R(t+h)$). For $d = 1$ (scalar y and r), $\beta(t+h)$ is a $(q+1) \times 1$ vector:

$$\beta(t+h) := \frac{P^-(t+h)_{:,1}}{S(t+h)}, \quad \text{where } S(t+h) = P^-(t+h)_{1,1} + R(t+h). \tag{11}$$

  Here:

  - $P^-(t+h)_{:,1}$ denotes the second column of the predicted covariance matrix $P^-(t+h)$ (assuming 0-indexed columns, this column corresponds to covariances with $X^{(1)}$). This term relates the uncertainty of the state to the uncertainty of the quantity being "measured" ($X^{(1)}$).

- $P^-(t+h)_{1,1}$ is the $(1,1)$-element of $P^-(t+h)$ (assuming 0-indexed rows/columns), representing the predicted variance of the first derivative, $m^{-(1)}(t+h)$.
- $S(t+h)$ is the **innovation covariance** (a scalar here), representing the total uncertainty in the innovation $r(t+h)$.

A larger gain means the filter relies more on the new measurement, while a smaller gain means it trusts its prediction more.

- **Updated Mean (Posterior Mean)**: The predicted mean is corrected using the innovation and the Kalman gain:

$$m(t+h) := m^-(t+h) + \beta(t+h)r(t+h). \tag{12}$$

(If r is d-dimensional, and β is $(q+1) \times d$, then it would be $\beta(t+h)r(t+h)$ if r is a column vector, or $\beta(t+h)r(t+h)^T$ if specific matrix/vector conventions are used by the paper, for $d = 1$ r is scalar so $r^T = r$). Assuming $d = 1$, $r(t+h)$ is scalar.

- **Updated Covariance (Posterior Covariance)**: The predicted covariance is also updated to reflect the reduction in uncertainty due to the measurement:

$$P(t+h) := P^-(t+h) - \beta(t+h)S(t+h)\beta(t+h)^T = P^-(t+h) - \frac{P^-(t+h)_{:,1}(P^-(t+h)_{:,1})^T}{P^-(t+h)_{1,1} + R(t+h)}. \tag{13}$$

This is a standard Kalman filter covariance update equation (Joseph form for scalar measurement).

The algorithm then repeats these prediction and update steps, advancing from $t = 0$ up to the final time T. The sequence $m^{(0)}(0), m^{(0)}(h), m^{(0)}(2h), \ldots, m^{(0)}(N_T h)$ (where $N_T = T/h$) provides the numerical solution to the ODE, along with a measure of uncertainty at each step via $P(nh)$.

To see how this algorithm works in practice, you can explore the Colab notebook here: Toy SDE Problem

3.3 Measurement Noise R (Section 2.3)

This subsection elaborates on the significance and origin of the measurement noise variance R.

The measurement noise R (or $R(t+h)$ in the context of the update step) is a critical parameter in the Kalman filter. It quantifies the believed uncertainty in the "measurement" $y(t+h) = f(m^{-(0)}(t+h))$ as an accurate representation of the true derivative $\dot{x}(t+h) = f(x^{(0)}(t+h))$. A well-chosen R is essential for the filter to perform optimally.

Two main sources contribute to the effective noise $R(t)$:

1. **Noise from imprecise knowledge of $x(t)$ (Discretization Error Proxy)**: The primary source of "noise" in $y(t+h)$ comes from the fact that $m^{-(0)}(t+h)$ (the predicted solution value) is only an estimate of the true solution value $x^{(0)}(t+h)$. The difference between $f(m^{-(0)}(t+h))$ and $f(x^{(0)}(t+h))$ depends on the error $e^{(0)}(t+h) = m^{-(0)}(t+h) - x^{(0)}(t+h)$. This error is related to the **discretization error** accumulated by the filter up to that point. This component of uncertainty generally increases with the step size h. So, R is implicitly linked to h.

2. **Noise from misspecified or imprecisely evaluated f**: In many real-world scenarios, the function f itself might not be known perfectly.

   - It might contain parameters that are estimated from data and are therefore uncertain.

- f could be computationally expensive to evaluate exactly, leading to the use of approximations or numerical methods that introduce their own errors.

The probabilistic framework allows, in principle, for such uncertainties in f to be incorporated into R. This is a distinct advantage over classical solvers, which typically assume f is known and evaluated perfectly.

In the paper's analysis, the focus is often on R as it depends on h through the numerical uncertainty in $x(t)$ (the first source). The paper analyzes the influence of noise R by assuming it is of order $p \in [1, \infty]$, meaning R scales proportionally to h^p ($R \propto h^p$).

- A large p implies R decreases rapidly as h gets smaller (meaning the "measurement" becomes very accurate relative to other errors).
- A small p (e.g., $p = 1$) implies R decreases slowly, or if p implies R is constant or grows, it means the measurement is fundamentally noisy.

Understanding this relationship helps determine whether the best strategy for improving accuracy is to reduce the step size h or to invest computational effort in obtaining a more precise evaluation of f (which would reduce R).

4 The Role of State Misalignments δ

This section introduces a subtle but fundamental concept in Gaussian ODE filtering: the potential discrepancy between the filter's estimates of derivatives and what those derivatives would be if strictly implied by the filter's estimate of the solution value and the ODE itself.

4.1 Analytical vs. Filter-Estimated Derivatives

If we knew the exact solution $x^{(0)}(t)$ to the ODE $x^{(1)}(t) = f(x^{(0)}(t))$, then all its higher time derivatives $x^{(i)}(t)$ would be uniquely determined. This can be seen by repeatedly differentiating the ODE with respect to time, using the chain rule. Let's define a sequence of functions $f^{(k)}$ that represent these ODE-implied derivatives:

- $f^{(-1)}(a) := a$ (This represents the function whose derivative we are looking at. For $x^{(0)}$, its "(-1) derivative" is just $x^{(0)}$ itself in some abstract sense for the recursion below).
- $f^{(0)}(a) := f(a)$ (This is the first derivative, $x^{(1)}(t) = f(x^{(0)}(t))$).
- For $i \geq 1$, if $x^{(i)}(t) = g(x^{(0)}(t))$ for some function g, then $x^{(i+1)}(t) = \frac{d}{dt} g(x^{(0)}(t)) = [\nabla_x g](x^{(0)}(t)) \cdot \frac{dx^{(0)}}{dt}(t) = [\nabla_x g](x^{(0)}(t)) \cdot f(x^{(0)}(t))$.
  So, $x^{(1)}(t) = f^{(1)}(x^{(0)}(t))$, and $x^{(2)}(t) = f^{(2)}(x^{(0)}(t))$, etc.

This analytical relationship suggests that if $m^{(0)}(t)$ is the filter's estimate of $x^{(0)}(t)$, then an "analytically desirable" estimate for the i-th derivative, let's call it $m^{(i)}_{\text{ideal}}(t)$, *ought to be* $m^{(i)}_{\text{ideal}}(t) = f^{(i)}_{\text{ODE-implied}}(m^{(0)}(t))$.

More formally, let $\Phi_0^{(i)}(a)$ denote the i-th time derivative (evaluated at some time $t' \geq 0$) of the **exact solution** to the original ODE, $x^{(1)}(t) = f(x^{(0)}(t))$, that starts with the initial condition $x(0) = a$. Then, if $m^{(0)}(t)$ is the filter's current estimate of the solution value at time t, the corresponding analytically consistent i-th derivative at that same time t would be $\Phi_t^{(i)}(m^{(0)}(t))$, where $\Phi_t^{(i)}$ is the i-th derivative of the flow starting from $m^{(0)}(t)$ *at time t* and evaluated at time t. Or, as the paper uses $\Phi_0^{(i)}(m^{(0)}(t))$ (Eq. (24) uses $m^{(0)}(nh)$), it means: if $m^{(0)}(t)$ *were* an initial condition at $t = 0$, $\Phi_0^{(i)}(m^{(0)}(t))$ would be its i-th derivative at time t. This sets up the reference for "perfect consistency."

However, the Gaussian ODE filter (the algorithm) *does not* explicitly compute or enforce this strict analytical relationship between $m^{(0)}(t)$ and $m^{(i)}(t)$ for $i \geq 1$. Doing so would generally be computationally intractable, as it would involve repeated symbolic differentiation and evaluation of complex Jacobian products. Instead, the filter:

1. Jointly models $x^{(0)}(t), x^{(1)}(t), \ldots, x^{(q)}(t)$ as components of the state vector $X(t)$.
2. Assumes these components evolve according to the prior SDE (Eq. (4)). This SDE has its own internal dynamics, e.g., that $X^{(i-1)}$ integrates to $X^{(i)}$ plus some noise, which is a statistical relationship, not strictly the ODE's analytical one for all derivatives.
3. Updates its belief about $X(t)$ using "measurements" derived from the ODE (specifically, $y = f(m^{-(0)})$ is used to inform the estimate of $X^{(1)}$).
4. The influence of this measurement on $X^{(0)}, X^{(2)}, \ldots, X^{(q)}$ is propagated through the Kalman gain and the covariance structure $P(t)$, which is shaped by the prior SDE.

As a consequence, the filter's computed mean estimates $m^{(i)}(t)$ for the derivatives (especially for $i \geq 2$) are a result of this statistical inference process. They depend not only on the ODE (via f) but also heavily on the chosen statistical model: the SDE prior (coefficients c_k, σ_j) and the assumed measurement noise R. Therefore, $m^{(i)}(t)$ will generally differ from the value that would be analytically derived solely from $m^{(0)}(t)$ and the ODE f.

4.2 Definition of State Misalignment $\delta^{(i)}(t)$

This "inevitable mismatch" or "discrepancy" between the filter's estimate of a derivative and the analytically consistent derivative (given the filter's estimate of the solution value) motivates the definition of the i-th state misalignment at time t. This is given by Equation (25) in the paper:

$$\delta^{(i)}(t) := \|m^{(i)}(t) - \Phi_0^{(i)}(m^{(0)}(t))\| \geq 0. \tag{14}$$

Let's clarify the terms:

- $m^{(i)}(t)$: This is the actual i-th component of the filter's mean state vector $m(t)$ at time t. It's the filter's working estimate for the i-th derivative of the solution.
- $m^{(0)}(t)$: This is the filter's estimate for the solution value $x^{(0)}(t)$ itself.
- $\Phi_0^{(i)}(m^{(0)}(t))$: This is the "ideal" or "analytically consistent" value for the i-th derivative at time t. It's what the i-th derivative *would be* at time t for an exact solution of the ODE that *started* with the initial condition $x(0) = m^{(0)}(t)$ (here $m^{(0)}(t)$ is treated as a hypothetical initial value at $t = 0$ for defining this reference trajectory).
- $\|\cdot\|$: Represents a suitable vector norm (e.g., Euclidean norm) that measures the magnitude of the difference.

Intuitively, $\delta^{(i)}(t)$ quantifies how far the filter's estimate of the i-th derivative, $m^{(i)}(t)$, has "drifted" from the value that would be perfectly consistent with its own estimate of the solution value, $m^{(0)}(t)$, if all relationships were dictated purely by the analytical structure of the ODE.

The misalignment $\delta^{(i)}(t)$ is zero if and only if $m^{(i)}(t) = \Phi_0^{(i)}(m^{(0)}(t))$.

- For $i = 0$: $\delta^{(0)}(t) = \|m^{(0)}(t) - \Phi_0^{(0)}(m^{(0)}(t))\|$. Since $\Phi_0^{(0)}(a)$ evaluated at time t is the solution $x_a(t)$ that started at a, this term is subtle. If $\Phi_0^{(0)}(m^{(0)}(t))$ means "the 0-th derivative (value) of the flow starting at $m^{(0)}(t)$ and evaluated at the same time t after evolving from $t = 0$ (where the initial condition was $m^{(0)}(t)$)", then it is just $m^{(0)}(t)$ if the flow map Φ_0 is properly defined for an arbitrary starting time. The paper states

$\delta^{(0)}(t) = 0$ because $\Phi_0^{(0)} = \mathrm{Id}$ (identity) likely in the sense that $\Phi_0^{(0)}(a)$ refers to the value a itself if no time evolution is considered, or that the reference is defined such that $m^{(0)}(t)$ is compared against itself. Let's assume for $i = 0$, the misalignment is defined to be zero by convention or specific definition of $\Phi_0^{(0)}$.

- For $i = 1$: $\delta^{(1)}(t) = \|m^{(1)}(t) - \Phi_0^{(1)}(m^{(0)}(t))\|$. Since $\Phi_0^{(1)}(a)$ (the first derivative of the flow starting at a) evaluated at time t is $f(x_a(t))$, where $x_a(t) = \Phi_0^{(0)}(a)$, this means $\Phi_0^{(1)}(m^{(0)}(t)) = f(\Phi_0^{(0)}(m^{(0)}(t)))$. If we are comparing $m^{(1)}(t)$ to $f(m^{(0)}(t))$ (the local ODE evaluation), then $\delta^{(1)}(t) = \|m^{(1)}(t) - f(m^{(0)}(t))\|$. This is zero if and only if the filter enforces $m^{(1)}(t) = f(m^{(0)}(t))$. In the Kalman filter update (Eq. (12) for the $m^{(1)}$ component), this only happens if the measurement noise $R = 0$ and the Kalman gain $\beta^{(1)}$ appropriately aligns these terms. If $R > 0$, the filter "hedges its bets" and $m^{(1)}(t)$ can differ from $f(m^{(0)}(t))$.

- For $i \geq 2$: $\delta^{(i)}(t)$ is generally non-zero. The filter does not directly use the complex analytical functions $\Phi_0^{(i)}$ (for $i \geq 2$) that would be needed to enforce this consistency. Its estimates $m^{(i)}(t)$ for higher derivatives are primarily shaped by the prior SDE dynamics (which propagate smoothness from $X^{(q)}$ downwards) and the indirect effect of the measurement $y = f(m^{-(0)})$ via the Kalman gain and covariance updates.

The existence of a non-zero misalignment $\delta^{(i)}(t) > 0$ is an important feature of these probabilistic solvers. It complicates the error analysis because the filter's behavior doesn't perfectly mirror a classical Taylor series expansion of the true solution. However, the paper aims to show that the influence of $\delta^{(i)}(t)$ can be bounded and, in a sense, is "swallowed by the Taylor remainder," meaning it doesn't fundamentally degrade the expected order of convergence of the method.

5 Auxiliary Bounds and Local Error Analysis (Section 5 & onwards)

This section marks the beginning of the formal error analysis, aiming to prove convergence rates for the probabilistic ODE filter. The strategy involves first establishing bounds on various intermediate quantities that arise in the algorithm and then using these to bound the local error (error committed in a single step).

5.1 Truncation Error $\varepsilon(nh)$

The primary quantity of interest in any numerical solver is the error it makes. For this filter, the **truncation error vector** (or global error vector) at a discrete time point $t_n = nh$ is defined as:

$$\varepsilon(nh) := m(nh) - x(nh) = \begin{pmatrix} m^{(0)}(nh) - x^{(0)}(nh) \\ \vdots \\ m^{(q)}(nh) - x^{(q)}(nh) \end{pmatrix} = \begin{pmatrix} \varepsilon^{(0)}(nh) \\ \vdots \\ \varepsilon^{(q)}(nh) \end{pmatrix}. \tag{15}$$

Here, $m(nh)$ is the filter's mean estimate of the augmented state (solution and its derivatives), and $x(nh)$ is the vector of the true solution and its true derivatives at time nh. The goal of the convergence analysis is to show that the norm of $\varepsilon(nh)$ (especially of its first component $\varepsilon^{(0)}(nh)$, the error in the solution value) tends to zero at a certain rate as the step size $h \to 0$.

5.2 Assumptions for Error Analysis

To mathematically analyze the filter's error, certain assumptions about the ODE problem and the filter's initialization are necessary.

- **Assumption 5.1** (Properties of f). *(Assumption 1 in the paper). This assumption generally states that the ODE function f is sufficiently smooth (e.g., Lipschitz continuous and its derivatives are bounded up to a certain order). This ensures that the true solution $x(t)$ exists, is unique, and is also sufficiently smooth. These properties are standard requirements for proving convergence of numerical ODE solvers.*

- **Assumption 5.2** (Initial Covariance Scaling). *(Assumption 2 in the paper). The entries of the initial covariance matrix $P(0)$ for the prior SDE satisfy:*

$$\|P(0)_{k,l}\| \leq K_0 h^{2q+1-k-l}, \quad \textit{for all } k, l \in \{0, \ldots, q\}, \tag{16}$$

where $K_0 > 0$ is a constant independent of h.

Intuition: This assumption ensures that the initial uncertainty prescribed by $P(0)$ is appropriately scaled with the step size h. The powers $2q + 1 - k - l$ are chosen such that the initial statistical uncertainty does not dominate the discretization uncertainty (which also depends on h) when h is small. For instance, the initial variance of the solution estimate $P(0)_{0,0}$ is $O(h^{2q+1})$, while for the q-th derivative $P(0)_{q,q}$ it's $O(h)$. This scaling reflects the idea that we might have less precise initial knowledge (or want to allow for more variance) for higher derivatives.

- **Assumption 5.3** (Initial Mean Error Scaling). *(Assumption 3 in the paper). The initial errors in the filter's mean estimate $m^{(i)}(0)$ for the i-th derivative satisfy:*

$$\|\varepsilon^{(i)}(0)\| = \|m^{(i)}(0) - x^{(i)}(0)\| \leq K_0 h^{q+1-i}, \quad \textit{for all } i \in \{0, \ldots, q\}. \tag{17}$$

Intuition: This assumption states that the initial guess $m(0)$ provided to the filter is already reasonably close to the true initial state $x(0)$ (and its true derivatives). The required accuracy of this initial guess depends on h and the order of the derivative i. For the solution value ($i = 0$), the initial error is assumed to be $O(h^{q+1})$, which is very small for small h. This is a strong assumption but is standard in local error analysis to isolate the error generated by one step of the method. It is weaker than assuming exact initialization ($\varepsilon^{(i)}(0) = 0$).

5.3 Key Lemmas and Theorem on Local Error

With these assumptions, the paper proceeds to bound crucial quantities.

Lemma 5.4 (Bound on Initial State Misalignment (Lemma 7)). *Under Assumption 5.1 (on f) and Assumption 5.3 (on initial mean error), for all $i \in \{0, \ldots, q\}$ and for all $h > 0$:*

$$\delta^{(i)}(0) \leq K h^{q+1-i}, \tag{18}$$

where K is a constant independent of h.

Significance: This lemma shows that if the filter starts with a good initial estimate $m(0)$ (as per Assumption 5.3), then the initial state misalignment $\delta(0)$ (the discrepancy between $m^{(i)}(0)$ and the analytically consistent derivative $\Phi_0^{(i)}(m^{(0)}(0))$) is also small and scales appropriately with h.

Proof Idea (from Eq. (33) in paper): The proof uses the triangle inequality:

$\delta^{(i)}(0) = \|m^{(i)}(0) - \Phi_0^{(i)}(m^{(0)}(0))\| \leq \|m^{(i)}(0) - x^{(i)}(0)\| + \|x^{(i)}(0) - \Phi_0^{(i)}(m^{(0)}(0))\|$.
The first term is $\|\varepsilon^{(i)}(0)\|$, bounded by $K_0 h^{q+1-i}$ (Assumption 5.3).
The second term, $\|x^{(i)}(0) - \Phi_0^{(i)}(m^{(0)}(0))\|$, represents how much the true i-th derivative at $t = 0$ differs from the i-th derivative of a trajectory that hypothetically started at $m^{(0)}(0)$. This difference can be related to $f^{(i-1)}(x^{(0)}(0)) - f^{(i-1)}(m^{(0)}(0))$ (using the ODE-implied derivative functions). By Lipschitz continuity of $f^{(i-1)}$ (which follows from Assumption 5.1), this difference is bounded by $L\|\varepsilon^{(0)}(0)\| \leq LK_0 h^{q+1}$. For $i < q+1$, h^{q+1-i} dominates h^{q+1}, so the overall bound for $\delta^{(i)}(0)$ is $O(h^{q+1-i})$.

The main result for the error committed in a single step is given by Theorem 8 of the paper.

Theorem 5.5 (Local Truncation Error (Theorem 8))**.** *Under Assumptions 5.1, 5.2, and 5.3, and for all sufficiently small $h > 0$, the error $\varepsilon(h) = m(h) - x(h)$ after one step of the filter (from $t = 0$ to $t = h$) satisfies:*

$$\|\varepsilon^{(0)}(h)\| \leq \|\varepsilon(h)\|_h \leq K_{LTE}[1 + \theta\|m^{(q)}(0)\|]h^{q+1}. \tag{19}$$

Here:

- $\varepsilon^{(0)}(h)$ is the error in the filter's estimate of the solution value $m^{(0)}(h)$ after one step.
- $\|\varepsilon(h)\|_h$ is the error vector $\varepsilon(h)$ measured in a specific weighted norm (defined by Eq. (19) in the paper, which typically scales components by powers of h to make them comparable in magnitude). The bound on $\|\varepsilon^{(0)}(h)\|$ is often the primary interest.
- K_{LTE} is a constant independent of h.
- The term $[1 + \theta\|m^{(q)}(0)\|]$ shows a dependency on the IOUP parameter θ (if $\theta > 0$) and the magnitude of the initial estimate of the q-th derivative. For the IBM prior ($\theta = 0$), this factor simplifies to 1.
- The crucial part is h^{q+1}. This indicates that the **local truncation error** in the solution estimate is of order $O(h^{q+1})$. This is the expected local order of accuracy for a numerical method that effectively utilizes information up to the q-th derivative (as the prior model does).

Theorem 5.5 also implies that the error in the i-th derivative estimate after one step, $\varepsilon^{(i)}(h)$, is bounded by order $O(h^{q+1-i})$. **Significance:** This theorem is fundamental. It confirms that, under appropriate conditions, a single step of the probabilistic filter behaves like a classical high-order numerical method in terms of the error it introduces. This is a prerequisite for proving global convergence.

Remark 5.6 (Implications of Theorem 5.5)**.**

- **Error Bounds for Derivatives**: A notable feature of this probabilistic approach, highlighted by Theorem 5.5, is that it provides explicit error bounds for the estimates of the derivatives ($m^{(i)}(h)$ for $i > 0$). Classical numerical ODE solvers (like Runge-Kutta) typically only provide an estimate for the solution $x(t)$ itself and do not directly estimate or provide error bounds for its derivatives. These derivative estimates and their error bounds can be valuable if the derivatives are needed for subsequent computations or analysis.
- **Influence and Complexity of the IOUP Prior ($\theta > 0$)**: The choice of prior (IBM vs. IOUP) affects the filter's behavior and the complexity of its analysis. For the IOUP prior ($\theta > 0$), the filter's mean prediction $m^-(t + h) = A(h)m(t)$ (derived from Eq. (8) in the paper for the SDE solution) inherently deviates from a pure Taylor series expansion of the solution (which is more closely related to the IBM prior where $\theta = 0$). Specifically, the

IOUP prior introduces terms proportional to θ that tend to "pull" the predicted solution towards a mean (often zero, depending on the specific SDE form). For example, the paper's Eq. (38) shows that for $q = 1$ (modeling $X^{(0)}, X^{(1)}$), the prediction $m^{-(0)}(nh+h)$ includes a term like $-\theta\frac{h^2}{2!}m^{(1)}(nh)$, in addition to the standard Taylor terms.

- This "pull" can be beneficial for ODEs whose solutions are inherently stable or decay (e.g., $\dot{x} = -x$), as the prior reinforces this behavior.
- However, it can be problematic for ODEs with diverging solutions (e.g., $\dot{x} = x$), as the IOUP prior might "fight against" the true dynamics of the ODE, potentially leading to larger errors or requiring stronger conditions for convergence.

Theorem 5.5 shows that this effect of θ is asymptotically negligible for *local* convergence (i.e., over a single step as $h \to 0$) because the θ-dependent terms in the local error bound are multiplied by h^{q+1}. However, these effects might accumulate and become significant for *global* convergence over the entire interval $[0, T]$. A rigorous global analysis for the IOUP case might necessitate stronger assumptions on the ODE function f (e.g., boundedness of f or its solution) to ensure that terms like $\|m^{(q)}(nh)\|$ remain controlled. Due to these added complexities, the paper restricts its initial global convergence analysis (in Section 7) to the simpler case of the IBM prior ($\theta = 0$).

6 Towards Global Analysis

The local error bounds established, particularly Theorem 5.5, are essential stepping stones for proving **global convergence rates**. Global convergence concerns how the total accumulated error $\varepsilon(T) = m(T) - x(T)$ at the final time T behaves as the step size $h \to 0$ (while the number of steps $N = T/h \to \infty$). Proving global convergence typically involves showing that the local errors do not accumulate too rapidly. The paper addresses this global analysis, focusing initially on the IBM prior. However, a detailed exposition of the global analysis is beyond the scope of this current document.

7 Conclusion

This document has provided a detailed walkthrough of the foundational concepts underpinning a probabilistic ODE solver based on Gaussian filtering, as presented in the research paper. We have explored:

- The formulation of a **prior model** for the ODE solution using Stochastic Differential Equations (SDEs), specifically the IBM and IOUP priors, which model the solution and its first q derivatives as a Gauss-Markov process.
- The iterative **Kalman-like filtering algorithm** (Ψ) that updates the probabilistic estimate of the state through prediction and update steps, incorporating information from the ODE function f as a "measurement."
- The role and interpretation of the **measurement noise** R, which accounts for discrepancies between $f(m^{-(0)})$ and the true derivative.
- The critical concept of **state misalignment** δ, which quantifies the deviation of the filter's derivative estimates from those strictly implied by its solution estimate and the ODE's analytical structure.
- The initial stages of the **error analysis**, including key assumptions on initial conditions and the derivation of a **local truncation error bound** (Theorem 5.5) showing the method achieves an error of order $O(h^{q+1})$ in a single step for the solution estimate.

This detailed understanding of the problem setup, the algorithm's mechanics, and the local error behavior forms a solid basis for appreciating the more advanced global convergence analysis presented in the full research paper and for understanding the unique advantages and considerations of such probabilistic numerical methods, particularly their ability to natively provide uncertainty quantification.